MSG No. 162.78 $P_c\bar{3}1m$ [Type IV, trigonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-3.m`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `2b`, site symmetry: `-3'.m`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,2,3,10,11,12,16,17,18,19,20,21]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[4,5,6,7,8,9,13,14,15,22,23,24]`

Table 3: Wyckoff site: `4c`, site symmetry: `3.2`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	`[1,2,3,4,5,6]`
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	`[7,8,9,10,11,12]`
3	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	`[13,14,15,16,17,18]`
4	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	`[19,20,21,22,23,24]`

Table 4: Wyckoff site: `4d`, site symmetry: `3.2'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{4}\right]$	`[1,2,3,16,17,18]`
2	$\left[\frac{1}{3}, \frac{2}{3}, \frac{3}{4}\right]$	`[4,5,6,13,14,15]`
3	$\left[\frac{2}{3}, \frac{1}{3}, \frac{3}{4}\right]$	`[7,8,9,22,23,24]`
4	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{4}\right]$	`[10,11,12,19,20,21]`

Table 5: Wyckoff site: `4e`, site symmetry: `3.m`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,10,11,12]`
2	$[0, 0, -z]$	`[4,5,6,7,8,9]`
3	$\left[0, 0, z + \frac{1}{2}\right]$	`[13,14,15,22,23,24]`
4	$\left[0, 0, \frac{1}{2} - z\right]$	`[16,17,18,19,20,21]`

Table 6: Wyckoff site: `6f`, site symmetry: `..2/m`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,5,7,11]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,6,8,12]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,4,9,10]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[13,17,19,23]
5	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[14,18,20,24]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[15,16,21,22]

Table 7: Wyckoff site: 6g, site symmetry: ..2'/m

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	[1,11,17,19]
2	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	[2,12,18,20]
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[3,10,16,21]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[4,9,15,22]
5	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	[5,7,13,23]
6	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	[6,8,14,24]

Table 8: Wyckoff site: 8h, site symmetry: 3..

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	[1,2,3]
2	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	[4,5,6]
3	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	[7,8,9]
4	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	[10,11,12]
5	$\left[\frac{1}{3}, \frac{2}{3}, z+\frac{1}{2}\right]$	[13,14,15]
6	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}-z\right]$	[16,17,18]
7	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}-z\right]$	[19,20,21]
8	$\left[\frac{2}{3}, \frac{1}{3}, z+\frac{1}{2}\right]$	[22,23,24]

Table 9: Wyckoff site: 12i, site symmetry: ..2

No.	position	mapping
1	$[x, -x, 0]$	[1,6]
2	$[x, 2x, 0]$	[2,4]
3	$[-2x, -x, 0]$	[3,5]
4	$[-x, x, 0]$	[7,12]
5	$[-x, -2x, 0]$	[8,10]
6	$[2x, x, 0]$	[9,11]
7	$\left[x, -x, \frac{1}{2}\right]$	[13,18]
8	$\left[x, 2x, \frac{1}{2}\right]$	[14,16]
9	$\left[-2x, -x, \frac{1}{2}\right]$	[15,17]
10	$\left[-x, x, \frac{1}{2}\right]$	[19,24]
11	$\left[-x, -2x, \frac{1}{2}\right]$	[20,22]
12	$\left[2x, x, \frac{1}{2}\right]$	[21,23]

Table 10: Wyckoff site: `12j`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\, -x,\, \frac{1}{4}\right]$	`[1,18]`
2	$\left[x,\, 2x,\, \frac{1}{4}\right]$	`[2,16]`
3	$\left[-2x,\, -x,\, \frac{1}{4}\right]$	`[3,17]`
4	$\left[x,\, 2x,\, \frac{3}{4}\right]$	`[4,14]`
5	$\left[-2x,\, -x,\, \frac{3}{4}\right]$	`[5,15]`
6	$\left[x,\, -x,\, \frac{3}{4}\right]$	`[6,13]`
7	$\left[-x,\, x,\, \frac{3}{4}\right]$	`[7,24]`
8	$\left[-x,\, -2x,\, \frac{3}{4}\right]$	`[8,22]`
9	$\left[2x,\, x,\, \frac{3}{4}\right]$	`[9,23]`
10	$\left[-x,\, -2x,\, \frac{1}{4}\right]$	`[10,20]`
11	$\left[2x,\, x,\, \frac{1}{4}\right]$	`[11,21]`
12	$\left[-x,\, x,\, \frac{1}{4}\right]$	`[12,19]`

Table 11: Wyckoff site: `12k`, site symmetry: `..m`

No.	position	mapping
1	$[x,\, 0,\, z]$	`[1,11]`
2	$[0,\, x,\, z]$	`[2,12]`
3	$[-x,\, -x,\, z]$	`[3,10]`
4	$[x,\, x,\, -z]$	`[4,9]`
5	$[-x,\, 0,\, -z]$	`[5,7]`
6	$[0,\, -x,\, -z]$	`[6,8]`
7	$\left[x,\, 0,\, z+\frac{1}{2}\right]$	`[13,23]`
8	$\left[0,\, x,\, z+\frac{1}{2}\right]$	`[14,24]`
9	$\left[-x,\, -x,\, z+\frac{1}{2}\right]$	`[15,22]`
10	$\left[x,\, x,\, \frac{1}{2}-z\right]$	`[16,21]`
11	$\left[-x,\, 0,\, \frac{1}{2}-z\right]$	`[17,19]`
12	$\left[0,\, -x,\, \frac{1}{2}-z\right]$	`[18,20]`

Table 12: Wyckoff site: `24l`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$[-y,\, x-y,\, z]$	`[2]`
3	$[-x+y,\, -x,\, z]$	`[3]`
4	$[x,\, x-y,\, -z]$	`[4]`
5	$[-x+y,\, y,\, -z]$	`[5]`
6	$[-y,\, -x,\, -z]$	`[6]`
7	$[-x,\, -y,\, -z]$	`[7]`
8	$[y,\, -x+y,\, -z]$	`[8]`
9	$[x-y,\, x,\, -z]$	`[9]`
10	$[-x,\, -x+y,\, z]$	`[10]`
11	$[x-y,\, -y,\, z]$	`[11]`
12	$[y,\, x,\, z]$	`[12]`

continued ...

Table 12

No.	position	mapping
13	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[13]
14	$\left[-y,\, x-y,\, z+\frac{1}{2}\right]$	[14]
15	$\left[-x+y,\, -x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[x,\, x-y,\, \frac{1}{2}-z\right]$	[16]
17	$\left[-x+y,\, y,\, \frac{1}{2}-z\right]$	[17]
18	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[18]
19	$\left[-x,\, -y,\, \frac{1}{2}-z\right]$	[19]
20	$\left[y,\, -x+y,\, \frac{1}{2}-z\right]$	[20]
21	$\left[x-y,\, x,\, \frac{1}{2}-z\right]$	[21]
22	$\left[-x,\, -x+y,\, z+\frac{1}{2}\right]$	[22]
23	$\left[x-y,\, -y,\, z+\frac{1}{2}\right]$	[23]
24	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[24]